Beyond Prototypes:
Longevity and Sustainability in Interactive Music Systems

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Trento 29 MARCH 2026





_A (classical) musician?
_An interaction designer?
_An activist for sustainable software or tech ethics?

Music

Music

Technology

Music

Technology

Research

Design

Music

Technology

Research

HCI

Design

Music

Technology

Research

HCI

Design

Music

Technology

Research

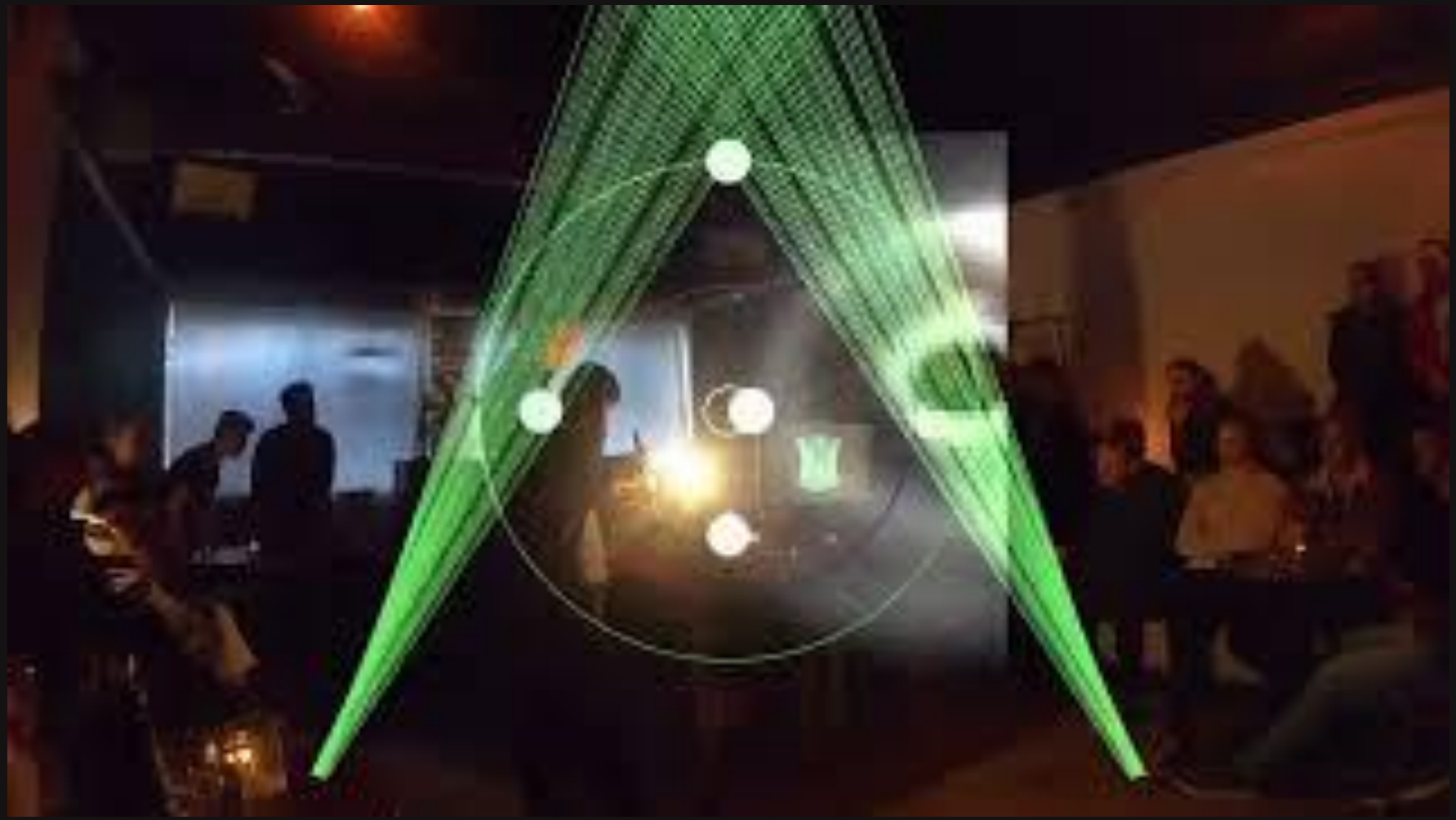
Performance

NIME

New Interfaces for Musical Expression

“started out as a workshop at the Conference on Human Factors in Computing Systems (CHI) in 2001”

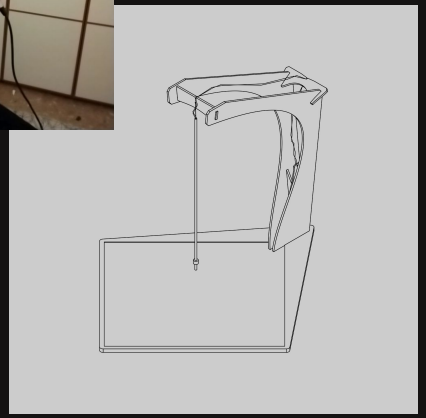

NME 2002 – Dublin








a
b
















Social
Environmental
Sustainability
Replicability
Longevity

_Cutural Diversity





Works with: Kenny Ma, Shiqi Li, Shumeng Zhang, Iuri Kusmin, O Al Kanawati

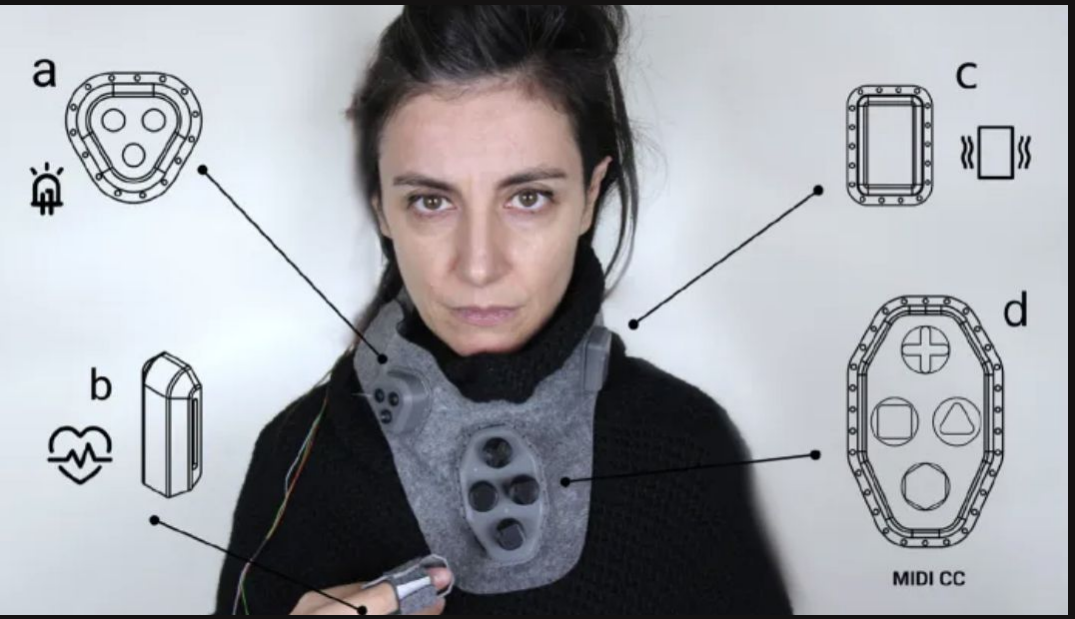








_Inclusivity



_With the singer and performer Eleonora Susanna

_Voice as expression of the body

_Diary and autoethnography

Check: https://eleonorasusanna.com/

_Cutural Diversity





Works with: Kenny Ma, Shiqi Li, Shumeng Zhang, Iuri Kusmin, O Al Kanawati









_Inclusivity: elderlie





Learning from HCI in and through Design

- sustainability in design, when the environmental issues are addressed as part of the material design of an artefact

- sustainability through design, when an artefact addresses environmental issues indirectly, focusing on changing ways of living social changes

(Mankov et al. CHI 2007)

As humans, we have a responsibility towards our environment and ecosystem.

As artists and researchers, we have the ability to give voice to climate change and to confront and question damaging privileges and habits, through the development of new practices, technologies and infrastructures that lessen our planetary impact.

As individuals, our impact is negligible but as a community, we can make a difference.

nime.org

Literature review 2001 - 2020

12 papers, we have identified three main concerns:

- Raising awareness of ecological and environmental issues
- Addressing environmental impact of music technology design
- Discussing ecological and environmental concerns within the context of political and cultural issues

NIME and the environment: toward a more sustainable NIME practice (Masu et al. NIME 2021)

Welcome

Copy

Info and resources for sustainable NIME research

To support sustainable practices within and beyond NIME as outlined in the NIME Conference Environmental Statement ↗, we have created this repository of information and green resources for environmental issues in NIME research. The repository is regularly updated with new entries, ideas, and suggestions for how we can lessen the ecological footprint of our individual and institutional practices. It is softly curated by the environmental officers, and we highly encourage everyone to submit requests for additions and edits. We hope that this will become a resource, not only for evaluating our own environmental impact, but also to spawn new ideas, designs, sonic works, and collaborations, with the potential for impact beyond the boundaries of NIME research as well.

> For more information on sustainable NIME research and practice, please see our 2021 NIME paper:
>
> - Raul Masu, Adam Pultz Melbye, John Sullivan, and Alexander Refsum Jensenius. 2021. NIME and the Environment: Toward a More Sustainable NIME Practice ↗. Proceedings of the International Conference on New Interfaces for Musical Expression. DOI: 10.21428/92fbeb44.5725ad8f ↗

Entries

		Research stage			
		Making	*Testing*	*Using*	*Disposing*
Resources	*Materials*				
	Consumables				
	Storage				
	Processing				
	Transport				

Reusing and Repurposing

27 papers 2020-22:

- Reusing Old NIMEs
- Updating Old NIMEs
- Complementing Old NIMEs
- Considering long-term engagement

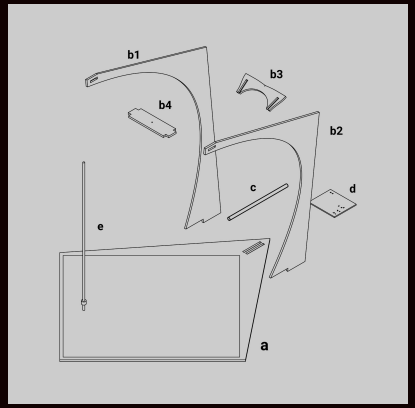
The O in NIME: Reflecting on the Importance of Reusing and Repurposing Old Musical Instruments (Masu et al. NIME 2022)

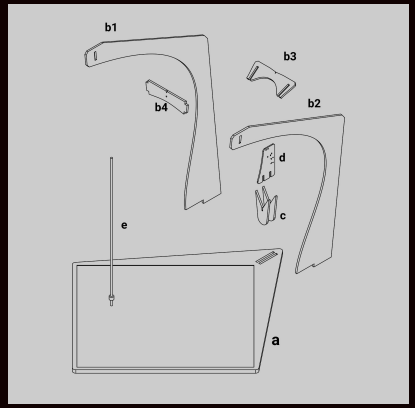
		Research stage			
		Making	*Testing*	*Using*	*Disposing*
Resources	*Materials*				
	Consumables				
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	Processing				
	Transport				

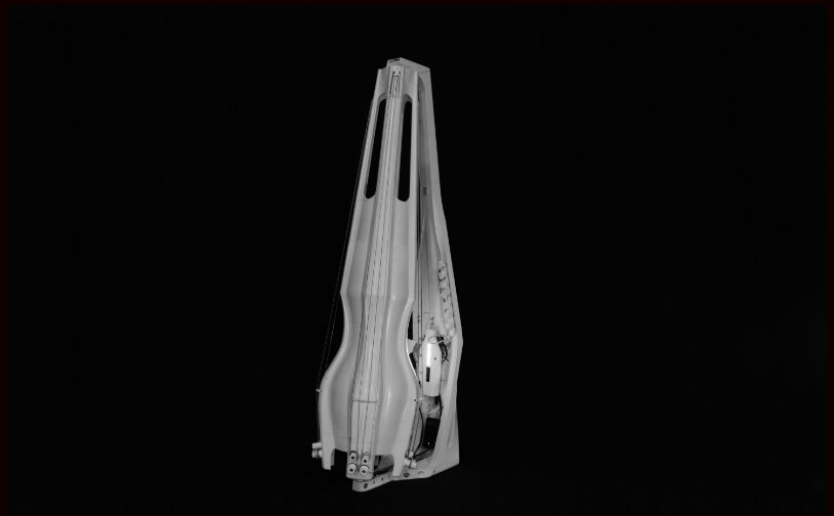
_Inscribe sustainability into practice

Research stage			
Making	*Testing*	*Using*	*Disposing*



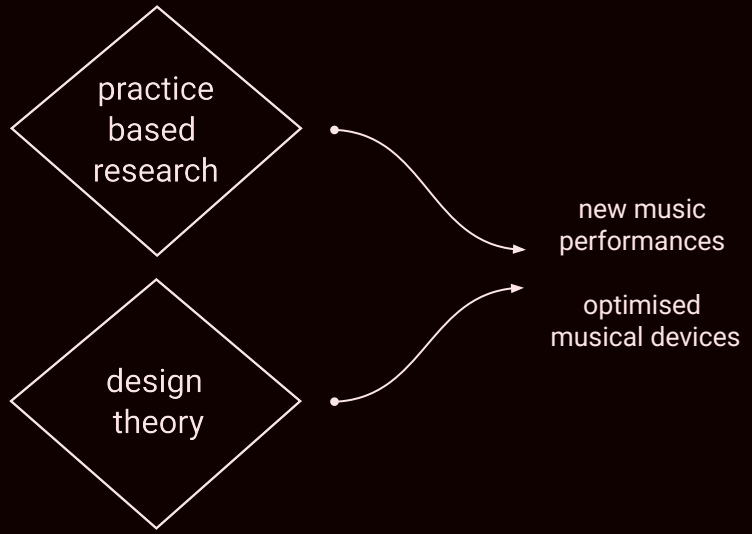






DMIs Longevity

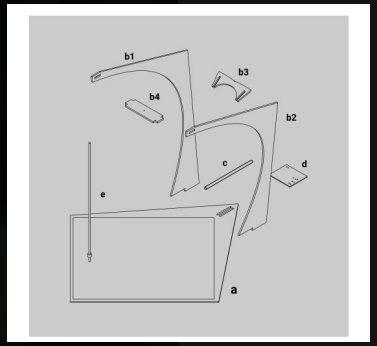
I have worked both on a theoretical level and with practical projects to foster DMIs sustainability, both by looking at design optimisation and documentation, and by reflecting on socio technical implications.
In this context I have collaborated with many musicians and researchers: Nicolò Merendino, Giacomo Lepri, Adam Pultz Melby, John Sullivan, Alexander Refsum Jensenius, Fabio Morreale



_Chowndolo



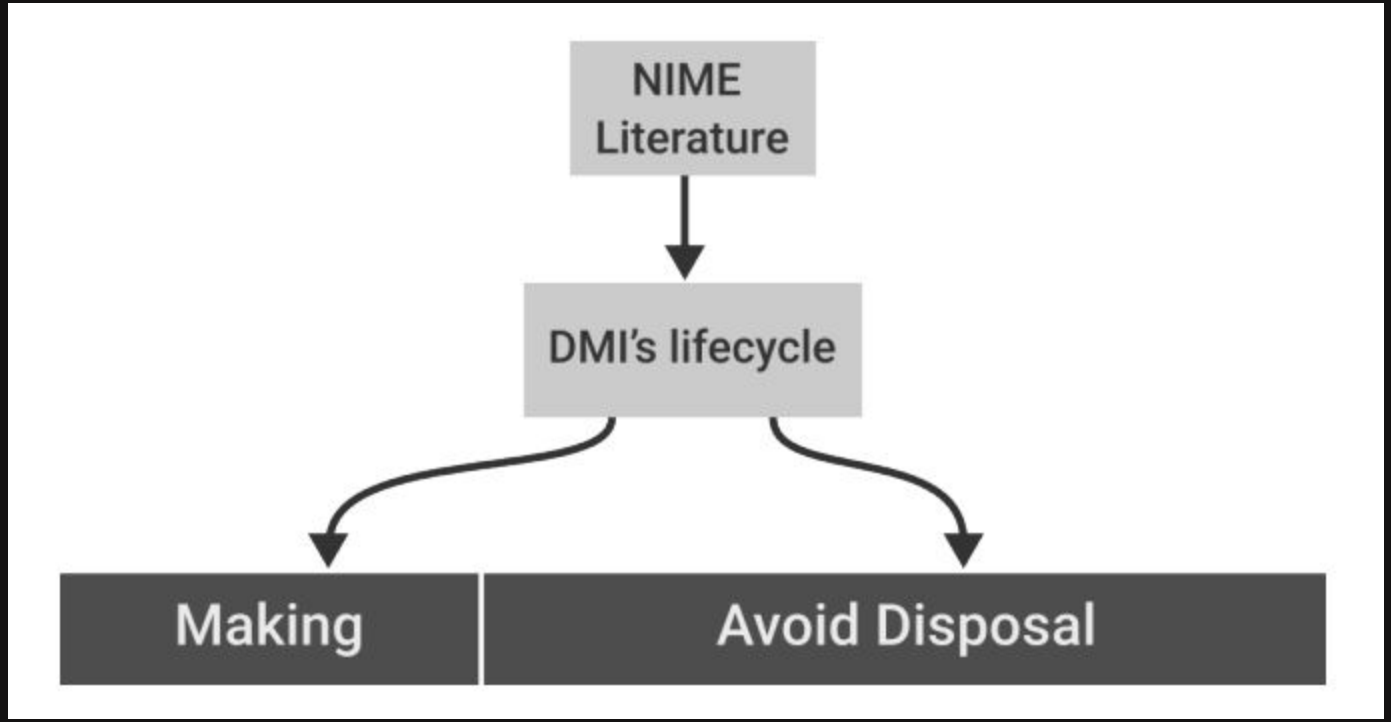
_Giacomo Lepri (main author)
_Nicolo Merendino (industrial designer)



Results (reflections)

_in design
    _Open Standards and Interoperability
    _Hardware Optimization and Durability
_through design
    _ Exposing the circuits
_proced based
    _Customizable Circuits
    _Documentation and Repository
    _Open Source

	Making			Avoid Disposal					
	1 Optimize fabrication process	2 Optimize logistics	3 Rely on recycled or sustainable materials	4 Document	5 Rely on modularity	6 Rely on Standard and interoperability	7 Consider affordability of reparation	8 Optimize and account for durability	9 Rely on FLOSS for fabrication
3D Printing									
Lasercutting									
CNC Milling									
Circuit Production									


NIME
Literature
DMI's lifecycle
Making
Avoid Disposal

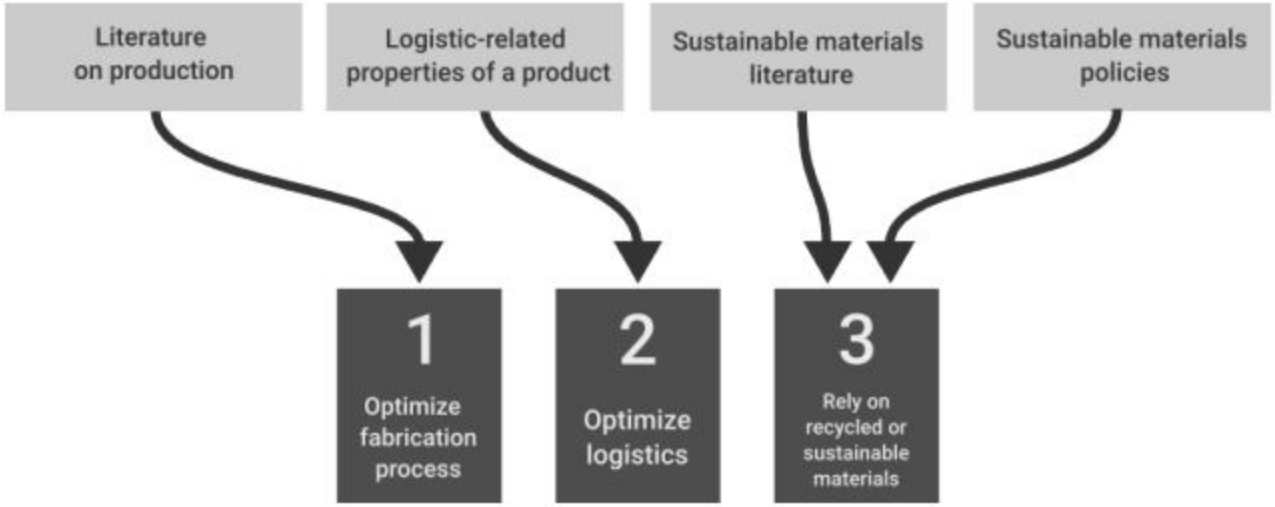
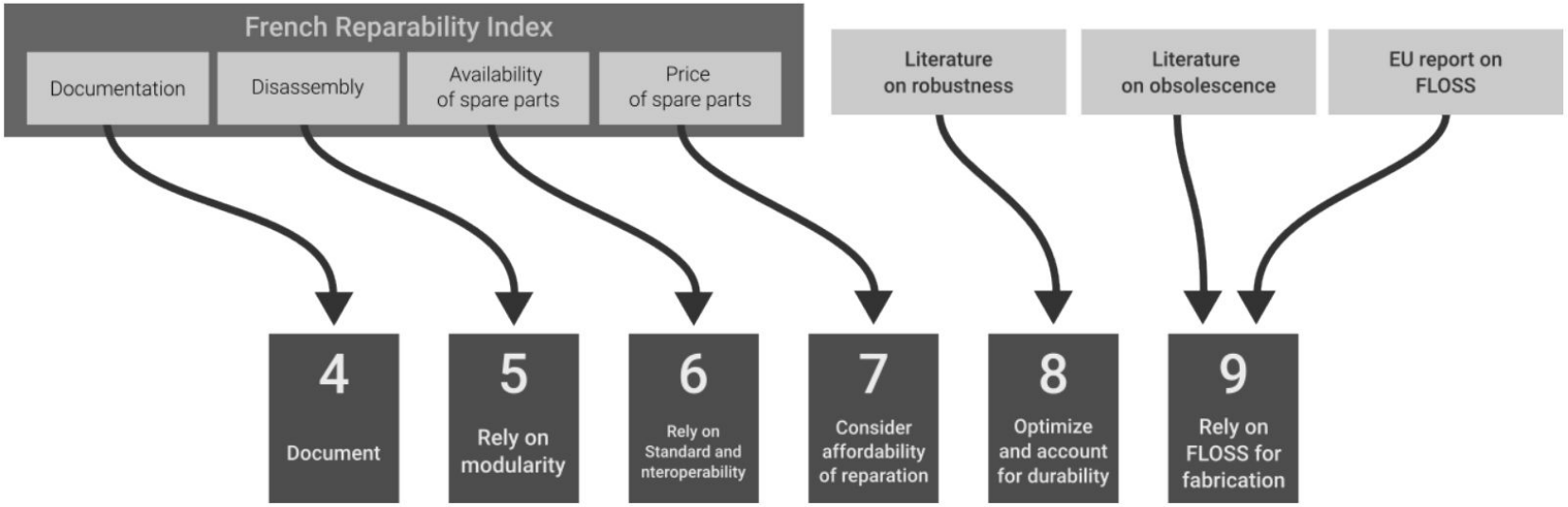
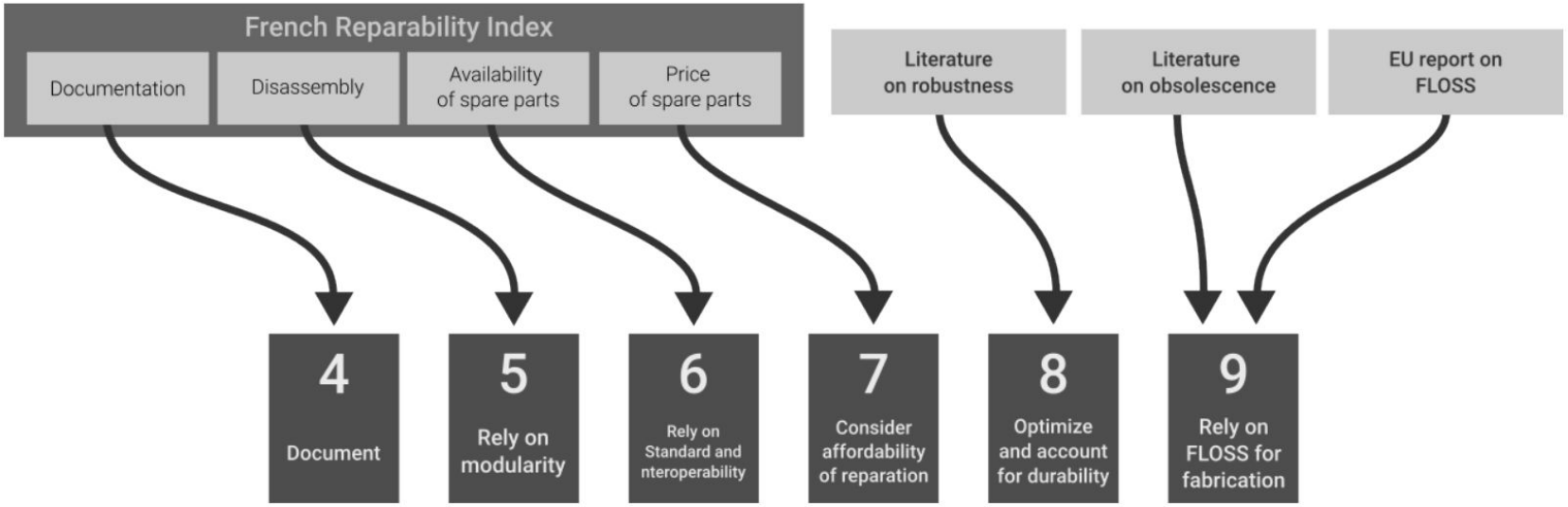
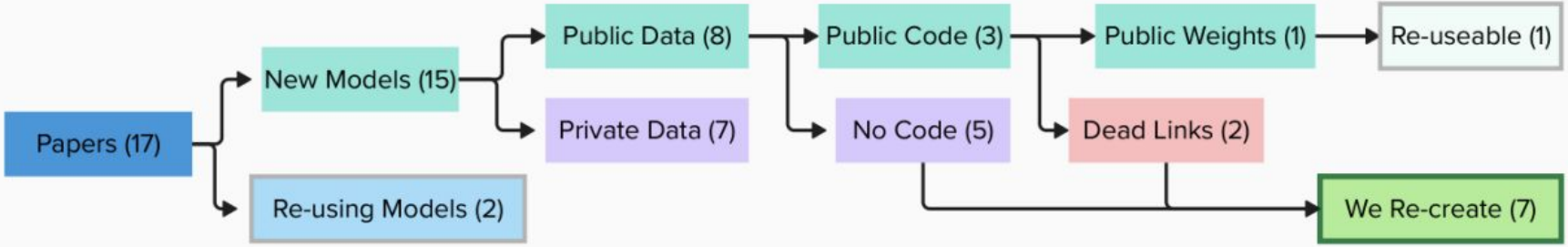


Figure 2: Diagram showing the rationale behind the Making related strategies


French Reparability Index
Documentation
Disassembly
Availability of spare parts
Price of spare parts
Literature on robustness
Literature on obsolescence
EU report on FLOSS
4
Document
5
Rely on modularity
6
Rely on Standard and nteroperability
7
Consider affordability of reparation
8
Optimize and account for durability
9
Rely on FLOSS for fabrication


French Reparability Index
Documentation
Disassembly
Availability of spare parts
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Clarks et al. Longevity of Deep Generative Models in NIME: Challenges and Practices for Reactivation

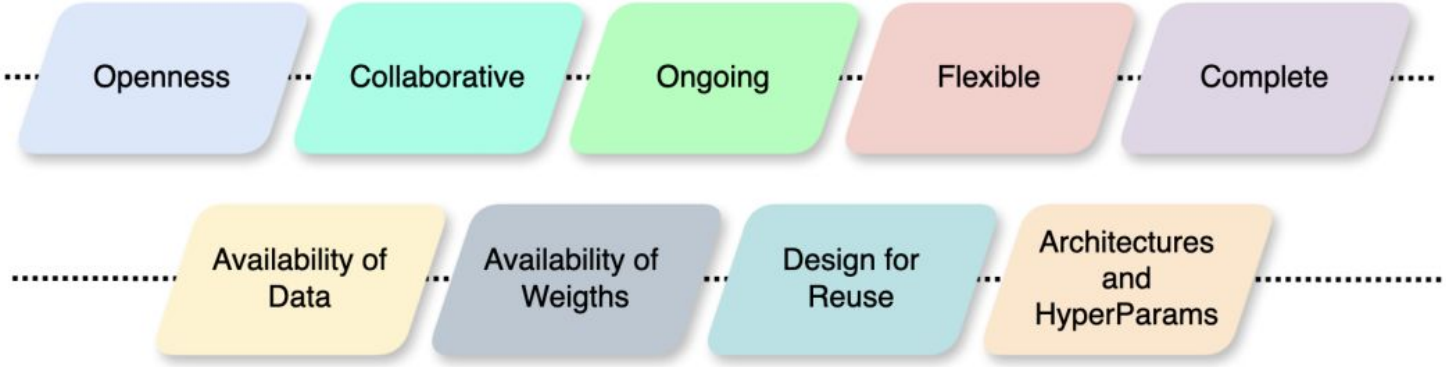
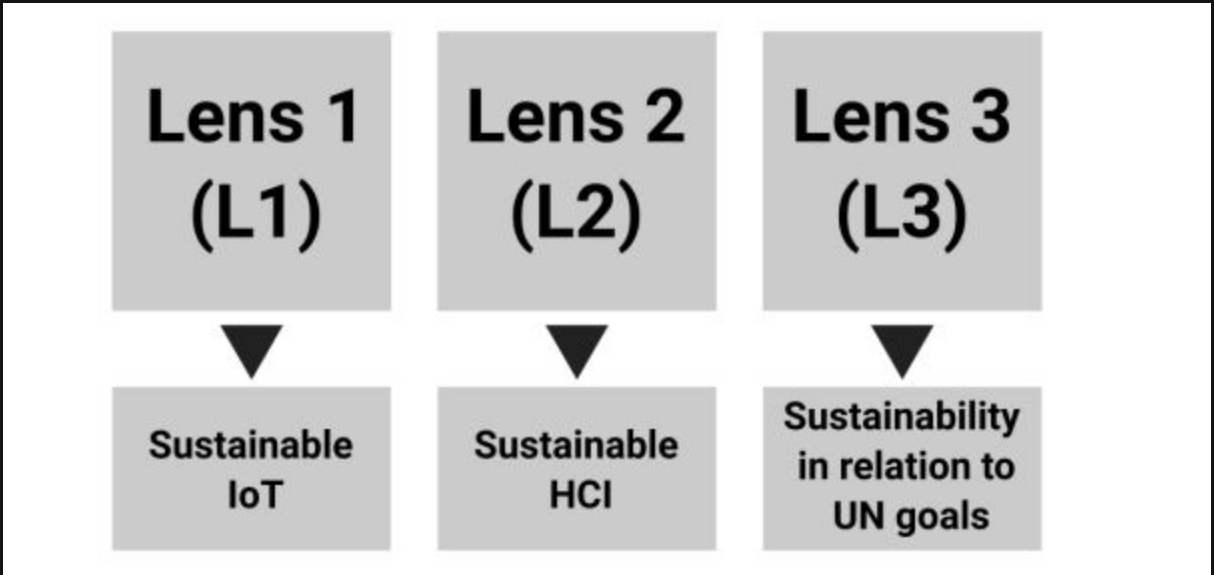
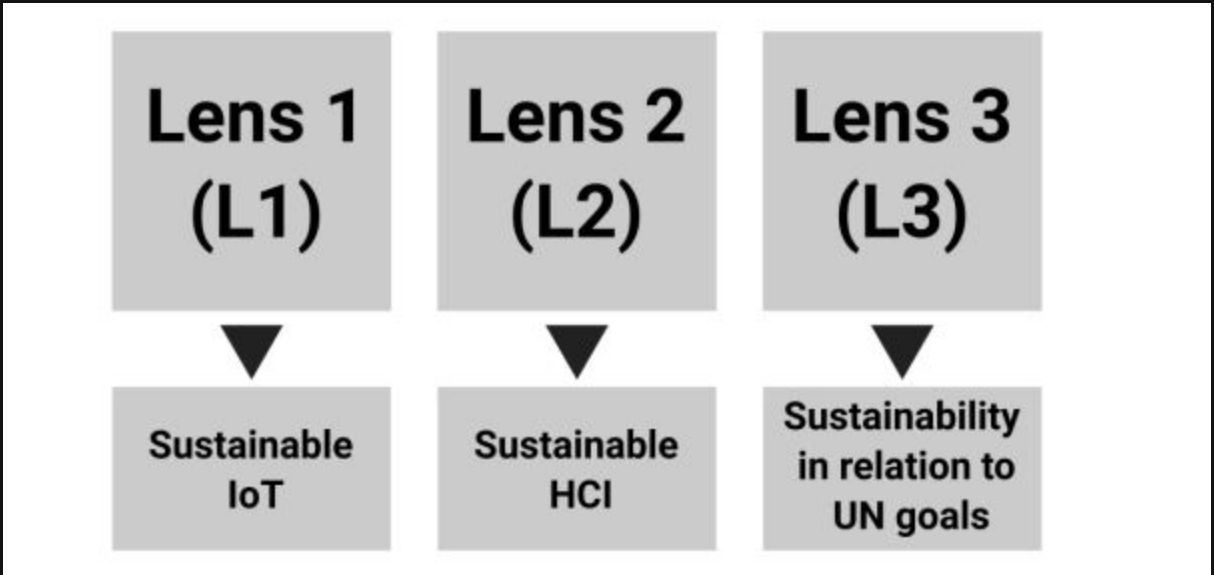


Figure 3: Bin's five suggestions for a critical history of NIME [3] over our four suggestions for preserving DGM NIMEs.

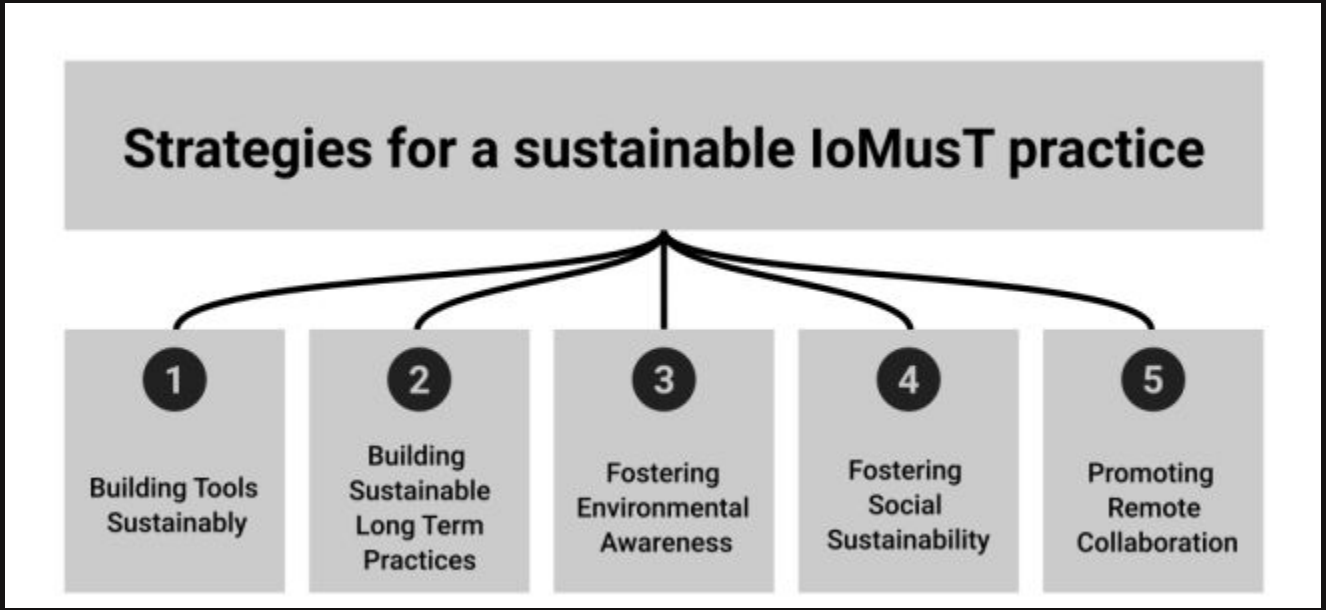
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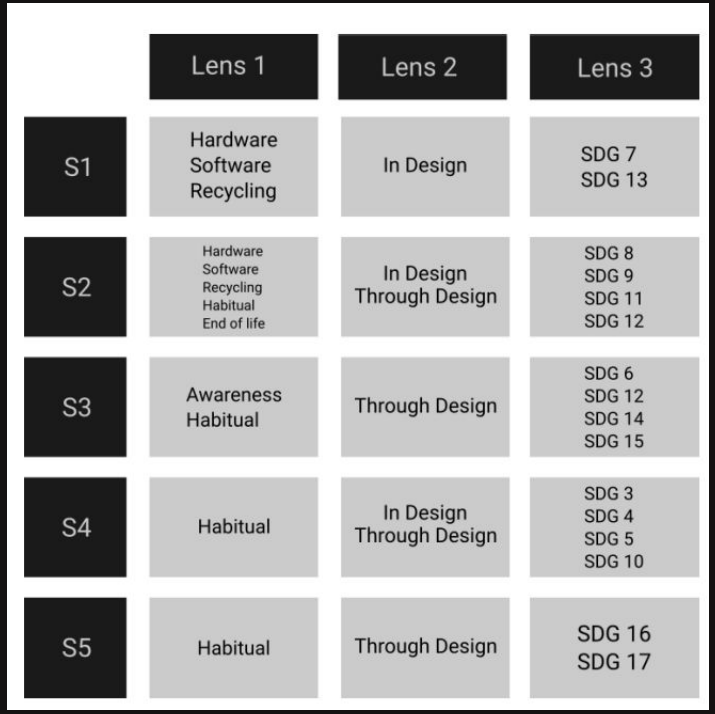
Sustainable Internet of Musical Things: Strategies to Account for Environmental and Social Sustainability in Network-Based Interactive Music Systems



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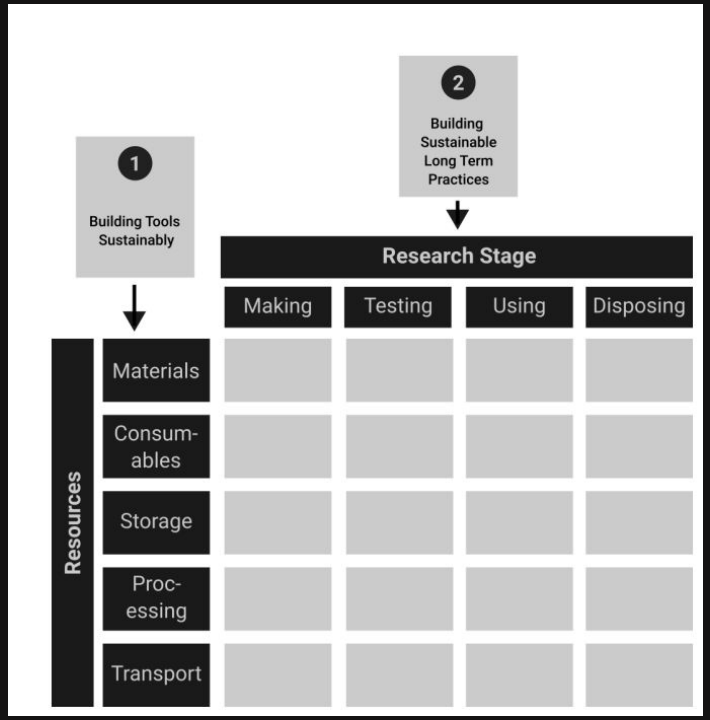


Sustainable Internet of Musical Things: Strategies to Account for Environmental and Social Sustainability in Network-Based Interactive Music Systems



	Lens 1	Lens 2	Lens 3
S1	Hardware Software Recycling	In Design	SDG 7 SDG 13
S2	Hardware Software Recycling Habitual End of life	In Design Through Design	SDG 8 SDG 9 SDG 11 SDG 12
S3	Awareness Habitual	Through Design	SDG 6 SDG 12 SDG 14 SDG 15
S4	Habitual	In Design Through Design	SDG 3 SDG 4 SDG 5 SDG 10
S5	Habitual	Through Design	SDG 16 SDG 17

Sustainable Internet of Musical Things: Strategies to Account for Environmental and Social Sustainability in Network-Based Interactive Music Systems



Sustainable Internet of Musical Things: Strategies to Account for Environmental and Social Sustainability in Network-Based Interactive Music Systems

_Inclusivity

_With the singer and performer Eleonora Susanna

_Voice as expression of the body

_Diary and autoethnography

Check: https://eleonorasusanna.com/

Reusing and Repurposing

- an “accelerated temporal logic of contemporary electronics [...] that insists on newness” (Devine 2019)
- a techno-utopian climate marked by “liberation through new technologies (that) mistakenly places technology, and not human agency, at the source of human history making” (Burkart 2012)
- a general academic climate that “privilege innovation and impact over actual content and substance” (Marquez-Borbon and P. Stapleton. 2015) and in which “counting matters” (Hayes and A. Marquez-Borbon 2020)
- a techno-cultural hegemony “rooted in the way (music) software is made, and those who make it” (Pardue and S. M. A. Bin 2022)
- a simplistic solution that reduces a complex space of “personally and artistically motivated challenges” (Morreale et al 2020), and of “politically negotiated demands” (Burkart and S. Leijonhufvud 2019) to neoliberal convenience.

Refs

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_Keep in touch

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